Bosch ADAS Training Data in Practice
博世智驾训练数据实战分享

From multimodal complexity
to a scalable data flywheel

DATA CENTER × TASK CENTER
Practice in Tencent Cloud

Three Problems, One Production Foundation

三个问题与一套数据底座

Efficient data use matters more than isolated storage optimization

UNIFIED MULTIMODAL DATA

统一多模态数据

Dense and sparse data must be managed together instead of becoming separate systems.

Dense 与 sparse 数据共同管理，避免形成相互割裂的体系。

EFFICIENT DATA USE

高效使用数据

Data should become train-ready with fewer copies, handoffs and alignment steps.

减少复制、交接和对齐，让数据交付后更快进入训练。

MODEL-DRIVEN FLYWHEEL

模型驱动的数据飞轮

Model failures should create the next data-production tasks and datasets.

由模型问题生成下一轮数据生产任务和目标数据集。

Lance + Ray: Data Centralization Meets Task Centralization

数据中心 + 任务中心：统一支撑数据生产

A shared foundation connects production, training and online evaluation

LANCE · DATA CENTER

One logical data layer manages multimodal datasets, versions and training access.

统一管理多模态数据集、版本及训练访问。

MODEL PROBLEM

RAY · TASK CENTER

One execution layer manages task submission, resources, status and aggregation.

统一管理任务提交、资源、状态及结果聚合。

DATA ↔ TASKS ↔ MODEL FEEDBACK

Manage Dense and Sparse Data as One Multimodal Dataset

Dense 与 sparse 共同构成一个可训练数据集

The delivery object becomes a train-ready dataset, not another intermediate handoff

DENSE DATA

Images, video and point clouds stay connected to the same dataset context.

图像、视频和点云始终关联在同一数据集上下文中。

SPARSE DATA

Labels, metadata and model outputs share identity, version and lineage.

标签、元数据和模型输出共享身份、版本与血缘。

TRAINING ACCESS

The delivered dataset can be consumed directly by training and evaluation.

交付后的数据集可以直接被训练和评测消费。

ONE DATASET · ONE VERSION · ONE PATH TO TRAINING

Reference: Lance documentation · immutable dataset versions and random access

Move from Local Validation to Cluster Execution with One Task Model

一套任务模型，从本地验证扩展到集群执行

Resource requests, execution status and results become visible and consistent

01

VALIDATE LOCALLY
本地验证

Estimate resources and verify logic in local mode.

在 local 模式验证逻辑并评估资源开销。

02

SUBMIT ONCE
一次提交

Send the validated job directly to the shared cluster.

将验证完成的任务直接提交到共享集群。

03

EXECUTE TRANSPARENTLY
透明执行

Schedule fine-grained tasks with explicit resources.

按明确资源需求调度细粒度任务。

04

AGGREGATE FAST
快速聚合

Collect distributed results and deliver them to the business.

聚合分布式结果并快速交付业务方。

Reference: Ray documentation · local-to-cluster execution and resource-aware scheduling

One Foundation Reduces Repeated Engineering and Maintenance

统一底座减少重复建设与长期维护成本

Infrastructure teams operate shared data and task primitives instead of business-specific stacks

ONE DATA FOUNDATION

统一数据底座

A shared multimodal data model replaces separate dense and sparse stacks.

以统一多模态模型替代 dense 与 sparse 两套体系。

ONE TASK FOUNDATION

统一任务底座

A shared execution model replaces business-specific schedulers and queues.

以统一执行模型替代业务自建调度器和队列。

LOWER MAINTENANCE

降低维护成本

Common interfaces, observability and operations reduce long-term cost.

统一接口、可观测性和运维方式降低长期成本。

STANDARDIZE THE BOTTOM LAYER; KEEP BUSINESS LOGIC FLEXIBLE.

底层能力标准化，保持灵活的业务逻辑。

A Transparent Process Turns Data Work into a Manageable Production Line

透明流程让数据工作成为可管理的生产线

Finer task and resource granularity improves utilization, traceability and delivery speed

TRANSPARENT PROCESS

过程透明

Every production stage has visible input, output, status and ownership.

FINER GRANULARITY

粒度更细

Tasks and resources are decomposed at the unit that can be scheduled efficiently.

CENTRALIZED DATA

数据集中

Intermediate and final products remain connected in one managed dataset.

HIGHER EFFICIENCY

效率更高

Less waiting and alignment shortens the path from request to training.

Model-Driven Data Production Loop

模型问题驱动的数据生产闭环

From model gaps to train-ready datasets and online evaluation

MODEL GAP
01

模型问题

Convert failures into data needs.

将失败案例转化为数据需求。

DATA MINING
02

数据挖掘

Discover target scenes and samples.

发现目标场景与候选样本。

AUTO-LABELING
03

自动标注

Generate scalable initial labels.

规模化生成初始标签。

HUMAN REPAIR
04

人工修复

Correct hard and long-tail cases.

修复困难及长尾场景。

TRAIN-READY DATASET
05

可训练数据集

Version and release qualified data.

对合格数据进行版本化交付。

TRAIN & EVALUATE
06

训练与评测

Train, evaluate and feed results back.

训练、评测并反馈新问题。

DATA CENTER • LANCE | TASK CENTER • RAY | 数据中心 + 任务中心

Mining Workloads: From Fragmented Scheduling to Transparent Cluster Execution

从分散任务调度到透明的集群执行

The same task moves from local resource validation to rapid distributed delivery

BEFORE

Scheduling patterns were inconsistent, resource estimates were isolated, and jobs often waited in queues.

调度模式不统一，资源开销各自评估，任务经常排队等待。

Result: high delivery pressure.

NOW WITH RAY

Validate resources locally, submit the same job to the cluster, and aggregate results through one visible workflow.

本地评估资源后直接提交集群，并在统一透明的流程中聚合结果。

Result: faster and more predictable delivery.

Data Production: From Lakehouse Handoff to Train-Ready Delivery

Training and online evaluation become direct consumers of the production dataset

BEFORE

Mining → auto-labeling → manual repair → lakehouse → downstream statistics → training alignment.

挖掘 → 自动标注 → 人工修复 → 湖仓 → 下游统计 → 训练对齐

Result: many handoffs before training.

结果：训练前仍需多次交接和重新对齐。

NOW WITH LANCE

Mining → labeling → repair → train-ready Lance dataset → training → online evaluation.

挖掘 → 标注 → 修复 → 可训练数据集 → 训练 → 在线评测

Result: delivery means training can start.

结果：数据达到交付标准即可启动训练。

TAKEAWAY

Let Model Problems Drive Data Production

模型问题 驱动 数据生产

LANCE

Centralize multimodal data and deliver train-ready datasets.

集中管理多模态数据，交付可直接训练的数据集。

RAY

Centralize tasks and scale validated workloads from local to cluster.

集中管理任务，让本地验证的作业平滑扩展到集群。

FLYWHEEL

Connect production, training and online evaluation through model feedback.

以模型反馈连接数据生产、模型训练与在线评测。

ONE DATA CENTER · ONE TASK CENTER · ONE CONTINUOUS LOOP